Introduction to Abstract Algebra III: Homework 3

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Victor Ostrik 11:00

Hashem A. Damrah
UO ID: 952102243

Exercise 40.33. Let E be an algebraically closed extension field of a field F. Show that the algebraic closure $\bar{F}_E$ of F in E is algebraically closed. (Applying this exercise to $\mathbb{C}$ and $\mathbb{Q}$, we see that the field of all algebraic numbers is an algebraically closed field.)

Solution. Let $\bar{F}_E$ be the algebraic closure of F in E. We want to show that $\bar{F}_E$ is algebraically closed, which means that every non-constant polynomial with coefficients in $\bar{F}_E$ has a root in $\bar{F}_E$. Let $f(x) \in \bar{F}_E[x]$ be a non-constant polynomial. Since $\bar{F}_E$ is an algebraic closure of F in E, every element of $\bar{F}_E$ is algebraic over F. Therefore, the coefficients of $f(x)$ are algebraic over F. Since E is an algebraically closed extension field of F, the polynomial $f(x)$ has a root α in E.

Now, we need to show that α is actually in $\bar{F}_E$. Since α is a root of $f(x)$, it is algebraic over $\bar{F}_E$. Moreover, since the coefficients of $f(x)$ are algebraic over F, α is also algebraic over F. Therefore, α is algebraic over F and belongs to E, which means that α must be in $\bar{F}_E$ (since $\bar{F}_E$ contains all elements of E that are algebraic over F). □

Exercise 40.35. Show that no finite field of odd characteristic is algebraically closed. (Actually, no finite field of characteristic 2 is algebraically closed either.) [Hint: By counting, show that for such a finite field F, some polynomial $x^2 - a$, for some $a \in F$, has no zero in F. See Exercise 32, Section 39.]

Solution. Assume that we have a finite field of odd characteristic, say F, with q elements. We want to show that F is not algebraically closed. Consider the polynomial $f(x) = x^2 - a$ for some $a \in F$. We will show that there exists an element $a \in F$ such that $f(x)$ has no root in F. Since F has q elements, there are q possible values for a. For each $a \in F$, the polynomial $f(x) = x^2 - a$ has at most 2 roots in F (since it is a quadratic polynomial). Therefore, the total number of roots of all such polynomials is at most $2q$. However, since there are q choices for a, the total number of roots can be at most $2q$. But we know that there are only q elements in F, so by the pigeonhole principle, there must be some element in F that is not a root of any of these polynomials. This means that there exists some $a \in F$ such that the polynomial $x^2 - a$ has no root in F, which implies that F is not algebraically closed. □

Exercise 41.3. Using the proof of Theorem 41.11, show that the regular 9-gon is not constructible.

Solution. Let $\theta = 40^\circ$. Then, $\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$. Letting $\alpha = \cos(\theta)$, we get:

$$4\alpha^3 - 3\alpha = -\frac{1}{2},$$

since $\cos(120^\circ) = -1/2$. Thus, α, the length of of one vertex of the polygon centered at the point of the polygon, is a zero of the polynomial:

$$f(x) = 8x^3 - 6x + 1.$$

By Theorem 28.12, $f(x)$ has no roots in the rationals, implying that $f(x)$ is irreducible over $\mathbb{Q}$. Thus, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Therefore, $120^\circ = 30^\circ \times 40^\circ$ cannot be trisected, implying that the regular 9-gon is inconstructible. □

Exercise 42.4. Find the number of primitive 8th roots of unity in GF(9).

Solution. The finite field GF(9) is an algebraic extension of GF(3) of degree 2, which means that GF(9) contains 9 elements. The multiplicative group of nonzero elements in GF(9) is cyclic of order 8. Therefore, there are $\varphi(8)$ primitive 8th roots of unity in GF(9), where φ is Euler's totient function. Computing, we find that $\varphi(8) = 4$. Therefore, there are 4 primitive 8th roots of unity in GF(9). □

Exercise 42.6. Find the number of primitive 15th roots of unity in GF(31).

Solution. Notice that 31 is a prime number, so GF(31) is a finite field with 31 elements. The multiplicative group of nonzero elements in GF(31) is cyclic of order 30. To find the number of primitive 15th roots of unity in GF(31), we need to find the number of elements in the multiplicative group that have order 15. The number of elements of order d in a cyclic group of order n is given by $\varphi(d)$ if d divides n, and 0 otherwise. Since 15 divides 30, we can compute $\varphi(15)$, which is 8. Therefore, there are 8 primitive 15th roots of unity in GF(31). □

Exercise 42.9. Let $\overline{\mathbb{Z}}_2$ be an algebraic closure of $\mathbb{Z}_2$, and let $\alpha, \beta \in \overline{\mathbb{Z}}_2$ be zeros of $x^3 + x^2 + 1$ and of $x^3 + x + 1$, respectively. Using the results of this section, show that $\mathbb{Z}_2(\alpha) = \mathbb{Z}_2(\beta)$.

Solution. First, we note that both $x^3 + x^2 + 1$ and $x^3 + x + 1$ are irreducible polynomials over $\mathbb{Z}_2$, as their roots aren't contained in $\mathbb{Z}_2$. Therefore, the degree of both fields $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ over $\mathbb{Z}_2$ is 3, which is prime. The order of both fields is $2^3 = 8$, which means that both fields are isomorphic to GF(8). By Lemma 42.8, the polynomial $x^8 - x$, has exactly 8 roots in GF(8), which are precisely the elements of GF(8). Since both α and β are roots of $x^8 - x$, they must be elements of GF(8). Therefore, both $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ are subfields of GF(8) that contain $\mathbb{Z}_2$. Since there is only one subfield of GF(8) that contains $\mathbb{Z}_2$ (which is GF(8) itself), we conclude that $\mathbb{Z}_2(\alpha) = \mathbb{Z}_2(\beta) = \mathrm{GF}(8)$. □

Exercise 42.11. Let F be a finite field of p^n elements containing the prime subfield $\mathbb{Z}_p$. Show that if $\alpha \in F$ is a generator of the cyclic group $\langle F^*, \cdot \rangle$, of nonzero elements of F, then $\deg(\alpha, \mathbb{Z}_p) = n$.

Solution. Let F be a finite field of p^n elements, and let $\alpha \in F$ be a generator of the cyclic group of nonzero elements of F. Since α generates the multiplicative group of F, the order of α is $p^n - 1$. The degree of α over $\mathbb{Z}_p$ is the smallest positive integer d such that $\alpha^d \in \mathbb{Z}_p$. Since $\mathbb{Z}_p$ has only p elements, we have that $\alpha^d = 1$ for some positive integer d. The order of α divides d, which means that $p^n - 1$ divides d. Therefore, the degree of α over $\mathbb{Z}_p$ is at least n. On the other hand, since $\alpha \in F$, we have that α is algebraic over $\mathbb{Z}_p$, and its minimal polynomial has degree at most n. Therefore, the degree of α over $\mathbb{Z}_p$ is at most n. Combining these two inequalities, we conclude that $\deg(\alpha, \mathbb{Z}_p) = n$. □

Exercise 42.12. Show that a finite field of p^n elements has exactly one subfield of p^m elements for each divisor m of n.

Solution. Let F be a finite field of p^n elements. Let m be a divisor n. First, we will show that there exists a subfield of F with p^m elements. Consider the polynomial $x^{p^m} - x$. This polynomial has at most p^m roots in F, and since F has p^n elements, it must have exactly p^m roots in F. Let K be the set of these roots. Then K is a subfield of F with p^m elements. □

Exercise 42.13. Show that $x^{p^n} - x$ is the product of all monic irreducible polynomials in $\mathbb{Z}_p[x]$ of a degree d dividing n.

Solution. Since $\mathbb{Z}_p$ is a field of prime characteristic, $x^{p^n} - x$ has p^n roots in the closure of $\mathbb{Z}_p$, $\overline{Z}_p$. Therefore, we can write it as follows:

$$f(x) = x^{p^n} - x = \prod_{\alpha \in \overline{\mathbb{Z}}_p} (x - \alpha).$$

Taking the derivative of $x^{p^n} - x$, we get:

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left(x^{p^n} - x\right) = p^n x^{p^n - 1} - 1.$$

Since $f(x)$ and $f'(x)$ share no roots, each α has a multiplicity of 1 in $f(x)$. □

Exercise 42.14. Let p be an odd prime.

(i) Show that for $a \in \mathbb{Z}$, where $a \not\equiv 0 \pmod p$, the congruence $x^2 \equiv a \pmod p$ has a solution in $\mathbb{Z}$ if and only if $a^{(p-1)/2} \equiv 1 \pmod p$. [Hint: Formulate an equivalent statement in the finite field $\mathbb{Z}_p$, and use the theory of cyclic groups.]

(ii) Using part (i), determine whether or not the polynomial $x^2 - 6$ is irreducible in $\mathbb{Z}_{17}[x]$.

Solution to (i). Assume that $x^2 \equiv a \pmod p$. Re-writing, we have $x^2 - a \equiv 0 \pmod p$. Let $f(x) = x^2 - a$. Then, there exists some $\alpha \in \mathbb{Z}_p$ such that $f(\alpha) = 0$. Since $\mathbb{Z}_p$ is a finite field, its multiplicative group of nonzero elements is cyclic. Let g be a generator of this group. Then, we can write $\alpha = g^m$, for some $m \in \mathbb{Z}_p$. Substituting back into the equation, we have $g^{2m} \equiv a \pmod p$. Since g is a generator, we can write $a = g^k$ for some $k \in \mathbb{Z}_p$. Therefore, we have $g^{2m} \equiv g^k \pmod p$, which implies that $2m \equiv k \pmod{p-1}$. This means that k must be even, and we can write $k = 2r$ for some $r \in \mathbb{Z}_p$. Substituting back, we have $a = g^{2r}$, which means that $a^{(p-1)/2} = (g^{2r})^{(p-1)/2} = g^{r(p-1)} = 1 \pmod p$.

Now, conversely, assume that $a^{(p-1)/2} \equiv 1 \pmod p$. Since the multiplicative group of nonzero elements in $\mathbb{Z}_p$ is cyclic, we can write $a = g^k$ for some $k \in \mathbb{Z}_p$. Then, we have $a^{(p-1)/2} = (g^k)^{(p-1)/2} = g^{k(p-1)/2} \equiv 1 \pmod p$. This implies that $k(p-1)/2$ is a multiple of $p-1$, which means that k must be even. Therefore, we can write $k = 2r$ for some $r \in \mathbb{Z}_p$. Substituting back, we have $a = g^{2r}$, which means that there exists some $\alpha = g^r$ such that $\alpha^2 = a$. Therefore, the congruence $x^2 \equiv a \pmod p$ has a solution in $\mathbb{Z}$. $\square$

Solution to (ii). Let $f(x) = x^2 - 6$. By part (i), there exists a solution in $\mathbb{Z}$ if and only if $6^8 \equiv 1 \pmod p$. But, computing the value, we find that $6^8 \equiv 16 \pmod{17}$. Therefore, there are no roots for the polynomial $f(x)$ in $\mathbb{Z}$, which means that $f(x)$ is irreducible. $\square$